

ESAT PRACTICE TEST SERIES

PHYSICS + BIOLOGY

LEVEL 0 • EASY

# Physics + Biology

## Mathematics 1 + Physics + Biology

Foundation-focused ESAT practice with more direct setup and shorter solution chains.

MODULE

**Mathematics 1**

MODULE

**Physics**

MODULE

**Biology**

# FULL SOLUTION BOOK

**ThrivingScholars**

81 preserved questions • three verified module keys

# Physics + Biology · Level 0 Easy

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

**Composition:** Mathematics 1 + Physics + Biology. Each module retains its original question order, answer and worked solution.

## Mathematics 1

Engineering Mock Test 1 · preserved Mathematics 1 module

|          |          |          |          |          |
|----------|----------|----------|----------|----------|
| Q1<br>G  | Q2<br>C  | Q3<br>B  | Q4<br>B  | Q5<br>E  |
| Q6<br>G  | Q7<br>D  | Q8<br>C  | Q9<br>B  | Q10<br>C |
| Q11<br>A | Q12<br>D | Q13<br>B | Q14<br>A | Q15<br>D |
| Q16<br>C | Q17<br>B | Q18<br>D | Q19<br>C | Q20<br>E |
| Q21<br>E | Q22<br>C | Q23<br>E | Q24<br>D | Q25<br>E |
| Q26<br>C | Q27<br>C |          |          |          |

## Physics

Engineering Mock Test 1 · preserved Physics module

|                        |                        |                        |                        |                        |
|------------------------|------------------------|------------------------|------------------------|------------------------|
| <b>Q1</b><br><b>F</b>  | <b>Q2</b><br><b>A</b>  | <b>Q3</b><br><b>D</b>  | <b>Q4</b><br><b>E</b>  | <b>Q5</b><br><b>G</b>  |
| <b>Q6</b><br><b>C</b>  | <b>Q7</b><br><b>D</b>  | <b>Q8</b><br><b>E</b>  | <b>Q9</b><br><b>D</b>  | <b>Q10</b><br><b>D</b> |
| <b>Q11</b><br><b>F</b> | <b>Q12</b><br><b>B</b> | <b>Q13</b><br><b>C</b> | <b>Q14</b><br><b>G</b> | <b>Q15</b><br><b>D</b> |
| <b>Q16</b><br><b>E</b> | <b>Q17</b><br><b>B</b> | <b>Q18</b><br><b>C</b> | <b>Q19</b><br><b>G</b> | <b>Q20</b><br><b>D</b> |
| <b>Q21</b><br><b>E</b> | <b>Q22</b><br><b>G</b> | <b>Q23</b><br><b>H</b> | <b>Q24</b><br><b>C</b> | <b>Q25</b><br><b>D</b> |
| <b>Q26</b><br><b>C</b> | <b>Q27</b><br><b>B</b> |                        |                        |                        |

## Biology

Preserved Biology Easy source set

|                        |                        |                        |                        |                        |
|------------------------|------------------------|------------------------|------------------------|------------------------|
| <b>Q1</b><br><b>C</b>  | <b>Q2</b><br><b>A</b>  | <b>Q3</b><br><b>H</b>  | <b>Q4</b><br><b>C</b>  | <b>Q5</b><br><b>E</b>  |
| <b>Q6</b><br><b>E</b>  | <b>Q7</b><br><b>C</b>  | <b>Q8</b><br><b>D</b>  | <b>Q9</b><br><b>B</b>  | <b>Q10</b><br><b>B</b> |
| <b>Q11</b><br><b>E</b> | <b>Q12</b><br><b>G</b> | <b>Q13</b><br><b>C</b> | <b>Q14</b><br><b>D</b> | <b>Q15</b><br><b>B</b> |
| <b>Q16</b><br><b>B</b> | <b>Q17</b><br><b>E</b> | <b>Q18</b><br><b>E</b> | <b>Q19</b><br><b>E</b> | <b>Q20</b><br><b>E</b> |
| <b>Q21</b><br><b>D</b> | <b>Q22</b><br><b>A</b> | <b>Q23</b><br><b>B</b> | <b>Q24</b><br><b>C</b> | <b>Q25</b><br><b>B</b> |
| <b>Q26</b><br><b>H</b> | <b>Q27</b><br><b>B</b> |                        |                        |                        |

# Mathematics 1

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 1 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

**M1-01**

Mock Test 1 - Mathematics 1

Answer **G****QUESTION 1**

If the 5-digit number  $1X2X3$  is divisible by 9, what is  $X$ ?

**A** 0**B** 1**C** 2**D** 3**E** 4**F** 5**G** 6**WORKED SOLUTION**

The digit sum is  $1 + X + 2 + X + 3 = 6 + 2X$ . For divisibility by 9,  $6 + 2X$  must be a multiple of 9. From the choices,  $X = 6$  gives 18.

**QUESTION 2**

If we define  $a \star b = a^2 - b^2$ , what is  $5 \star 4$ ?

A 1

B 7

C 9

D 16

E 25

F 41

**WORKED SOLUTION**

Apply the definition directly:  $5 \star 4 = 5^2 - 4^2 = 25 - 16 = 9$ .

**QUESTION 3**

Jack tosses two fair dice. What is the probability that the sum of the two values is no less than 9?

Recall that

$$\text{Probability} = \frac{\text{Number of valid cases}}{\text{Number of all possible cases}}.$$

**A**  $\frac{1}{4}$

**B**  $\frac{5}{18}$

**C**  $\frac{1}{3}$

**D**  $\frac{5}{12}$

**E**  $\frac{2}{3}$

**WORKED SOLUTION**

For two dice, sums at least 9 are 9, 10, 11, 12, giving  $4 + 3 + 2 + 1 = 10$  outcomes. So the probability is  $10/36 = 5/18$ .

**M1-04**

Mock Test 1 - Mathematics 1

Answer **B**

**QUESTION 4**

How many 4-digit integers  $\overline{abcd}$  satisfy that

- 1 The thousands digit  $a$  and the tens digit  $c$  are even numbers;
- 2 The hundreds digit  $b$  and the unit digit  $d$  are odd numbers.
- 3 This number is divisible by 5?

**A** 80

**B** 100

**C** 125

**D** 250

**E** 500

**WORKED SOLUTION**

Since the number is divisible by 5, the final digit must be 5 because  $d$  is odd.

Then  $a \in \{2, 4, 6, 8\}$ ,  $b$  has 5 choices, and  $c$  has 5 choices, giving

$$4 \cdot 5 \cdot 5 \cdot 1 = 100.$$

**M1-05**

Mock Test 1 - Mathematics 1

Answer **E**

**QUESTION 5**

If  $\frac{m}{10} = \frac{6}{n} = \frac{2}{5}$ , what is  $m + n$ ?

**A** 10

**B** 13

**C** 14

**D** 16

**E** 19

**WORKED SOLUTION**

From  $m/10 = 2/5$ ,  $m = 4$ . From  $6/n = 2/5$ ,  $30 = 2n$ , so  $n = 15$ . Hence  $m + n = 19$ .

**QUESTION 6**

Nancy has 100 marbles. She gives 20% of them to her friend Pedro. Then Nancy gives 25% of what is left to another friend, Ben. Finally, Nancy gives 10% of what is now left in the bag to her brother Jimmy. What percentage of her original bag of marbles does Nancy have left for herself?

**A** 20

**B** 30

**C**  $33\frac{1}{3}$

**D** 38

**E** 45

**F** 50

**G** 54

**H** 60

**WORKED SOLUTION**

Track what remains: after 20% is given away, 80 remain; after giving 25% of 80, 60 remain; after giving 10% of 60, 54 remain.

**QUESTION 7**

What is the sum of all positive 2-digit integers that have a remainder of 1 when divided by 9 and a remainder of 2 when divided by 5?

**A** 29

**B** 56

**C** 84

**D** 119

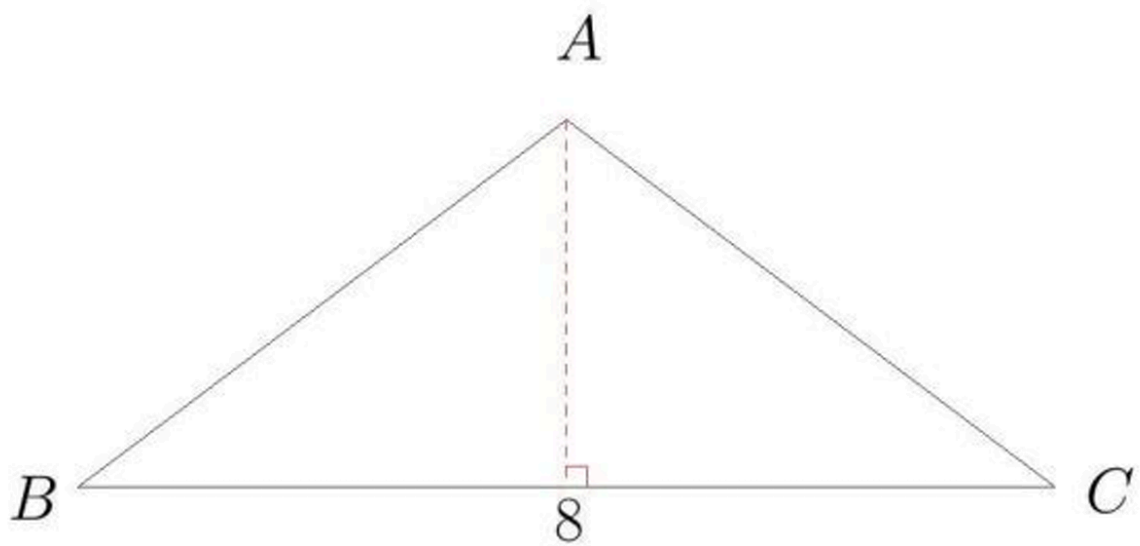
**E** 164

**WORKED SOLUTION**

The numbers satisfying both conditions are 37 and 82: each is 1 more than a multiple of 9 and 2 more than a multiple of 5. Their sum is 119.

**QUESTION 8**

$\triangle ABC$  is an isosceles triangle.  $AB = AC$ . If  $BC = 8$  and the area of  $\triangle ABC$  is 12, what is the perimeter of  $\triangle ABC$ ?



**A** 12

**B** 16

**C** 18

**D** 19

**E** 20

**WORKED SOLUTION**

The height is  $h$  where  $\frac{1}{2} \cdot 8 \cdot h = 12$ , so  $h = 3$ . The altitude bisects the base, giving half-base 4, so each equal side is  $\sqrt{3^2 + 4^2} = 5$ . Perimeter  $= 8 + 5 + 5 = 18$ .

**QUESTION 9**

What is the smaller angle between the hour hand and minute hand of a clock at 5:50 pm?

**A**  $100^\circ$

**B**  $125^\circ$

**C**  $150^\circ$

**D**  $155^\circ$

**E**  $165^\circ$

**WORKED SOLUTION**

At 5:50, the minute hand is at  $300^\circ$ . The hour hand is at  $5 \cdot 30 + 50 \cdot 0.5 = 175^\circ$ . The smaller angle is  $300 - 175 = 125^\circ$ .

**QUESTION 10**

The average age of 5 people in a room is 40 years. A 36-year-old person leaves the room. What is the average age of the four remaining people?

**A** 39

**B** 40

**C** 41

**D** 43

**E** 45

**WORKED SOLUTION**

The original total age is  $5 \cdot 40 = 200$ . After the 36-year-old leaves, the total is 164, so the new average is  $164/4 = 41$ .

**M1-11**

Mock Test 1 - Mathematics 1

Answer **A**

**QUESTION 11**

In how many ways can we write 127 as the sum of two prime numbers?

**A** 0

**B** 1

**C** 2

**D** 3

**E** 4

**WORKED SOLUTION**

An odd number written as a sum of two primes must use 2 plus an odd prime. Here  $127 = 2 + 125$ , but 125 is not prime, so there are no ways.

**QUESTION 12**

If for some positive integer  $n$ ,

$$\left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{3}\right) \left(1 + \frac{1}{4}\right) \cdots \left(1 + \frac{1}{n}\right) = 50,$$

what is  $n$ ?

**A** 20

**B** 39

**C** 65

**D** 99

**E** 100

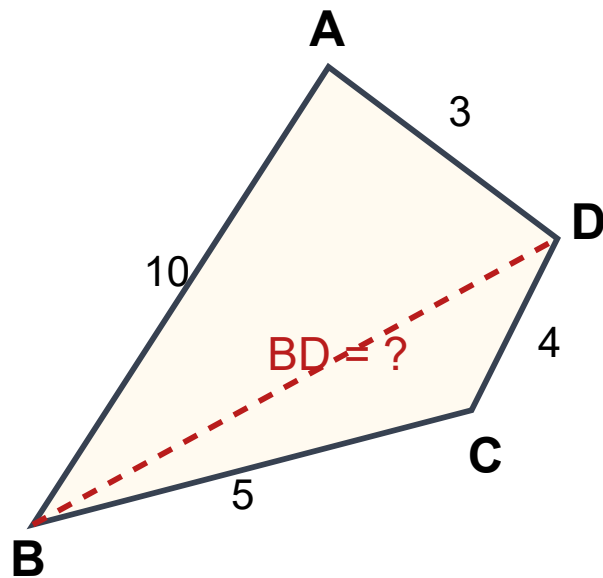
**WORKED SOLUTION**

Each factor is  $1 + \frac{1}{k} = \frac{(k+1)}{k}$ . The product telescopes:

$$\frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n} = \frac{n+1}{2} = 50. \text{ Thus } n = 99.$$

**QUESTION 13**

In quadrilateral  $ABCD$ ,  $AB = 10$ ,  $BC = 5$ ,  $CD = 4$  and  $AD = 3$ . If the length of  $BD$  is an integer, what is  $BD$ ?



**A** 7

**B** 8

**C** 9

**D** 10

**E** there is not enough information

#### WORKED SOLUTION

Use the triangle inequality in the two triangles that share  $BD$ . For  $\triangle ABD$ ,  $|10 - 3| < BD < 10 + 3$ , so  $7 < BD < 13$ . For  $\triangle BCD$ ,  $|5 - 4| < BD < 5 + 4$ , so  $1 < BD < 9$ . The only integer satisfying both is  $BD = 8$ .

**QUESTION 14**

What is the unit digit of  $7^{2020}$ ? Here

$$7^n = \underbrace{7 \times 7 \times 7 \times \dots \times 7}_n$$

(Hint: check unit digits of  $7$ ,  $7^2$ ,  $7^3$ , ... and find the pattern.)

**A** 1

**B** 3

**C** 5

**D** 7

**E** 9

**WORKED SOLUTION**

The units digits of powers of 7 cycle as 7, 9, 3, 1. Since 2020 is divisible by 4, the units digit is the fourth term of the cycle, 1.

**QUESTION 15**

Here  $x$ ,  $y$ ,  $z$  and  $u$  represent four different numbers.

$$\begin{array}{r} x \quad y \quad x \quad y \\ + z \quad x \quad - z \quad x \\ \hline u \quad x \quad \quad x \end{array}$$

$$x + y + z + u =$$

**A** 12

**B** 14

**C** 16

**D** 18

**E** 20

#### WORKED SOLUTION

From the subtraction,  $(10x + y) - (10z + x) = x$ , so  $8x + y = 10z$ . From the addition,  $(10x + y) + (10z + x) = 10u + x$ , so  $10x + y + 10z = 10u$ . The distinct digit solution is  $(x, y, z, u) = (5, 0, 4, 9)$ , giving 18.

**M1-16**

Mock Test 1 - Mathematics 1

Answer C

#### QUESTION 16

Billy, a seasonal worker in the town of Cowra, collected 8 buckets of cherries on his first day. Each day after that he increased the number of buckets he picked by 2 buckets per day. According to this pattern, how many buckets will he collect in the first 20 days?

**A** 308

**B** 500

**C** 540

**D** 580

**E** 600

**WORKED SOLUTION**

The buckets form an arithmetic sequence: 8, 10, 12, ... . The 20th term is  $8 + 19 \cdot 2 = 46$ . Sum =  $\frac{20}{2}(8 + 46) = 540$ .

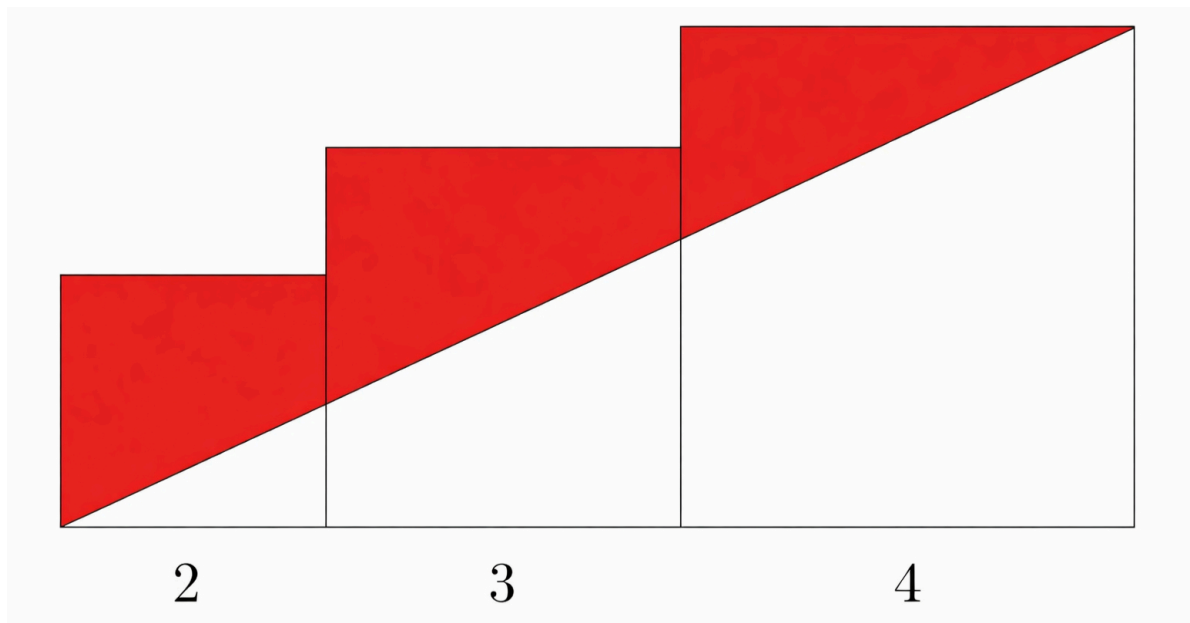
**M1-17**

Mock Test 1 - Mathematics 1

Answer **B**

**QUESTION 17**

Three squares are lined up horizontally as shown. Their side lengths are 2, 3 and 4 respectively. A straight line is drawn from the top right corner of the largest square to the bottom left hand corner of the smallest square. What is the area of the shaded region?



**A** 10

**B** 11

**C** 13

**D** 16

**E** 18

**WORKED SOLUTION**

The total area of the three squares is  $2^2 + 3^2 + 4^2 = 29$ . The line cuts off one large triangle of base 9 and height 4, area 18. Shaded area =  $29 - 18 = 11$ .

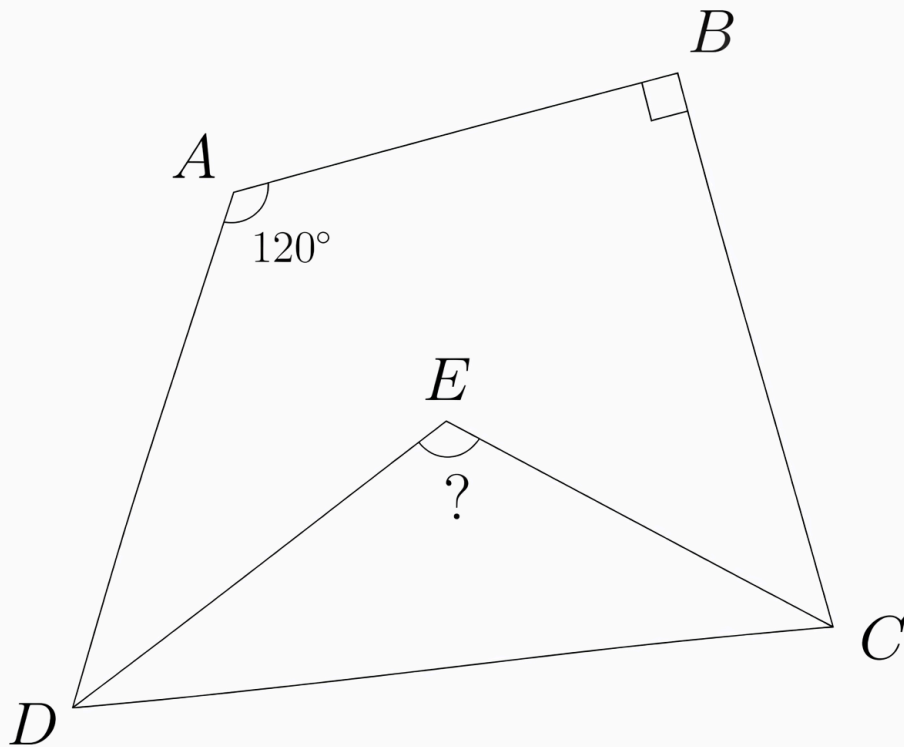
**M1-18**

Mock Test 1 - Mathematics 1

Answer D

**QUESTION 18**

In quadrilateral  $ABCD$ ,  $\angle A = 120^\circ$  and  $\angle B = 90^\circ$ .  $ED$  and  $EC$  bisect  $\angle ADC$  and  $\angle BCD$  respectively. What is  $\angle E$ ?



**A** 90

**B** 95

**C** 100

**D** 105

**E** 110

**WORKED SOLUTION**

In the quadrilateral,  $\angle C + \angle D = 360^\circ - 120^\circ - 90^\circ = 150^\circ$ . Since  $EC$  and  $ED$  bisect those angles, triangle  $DCE$  has angles at  $C, D$  summing to  $75^\circ$ .  
Therefore  $\angle E = 180^\circ - 75^\circ = 105^\circ$ .

**M1-19**

Mock Test 1 - Mathematics 1

Answer **C**

**QUESTION 19**

In a mathematics contest with 10 problems, a student gains 6 points for a correct answer and loses 1 point for an incorrect answer. If Olivia answered every problem and her score was 32, how many correct answers did she have?

**A** 4

**B** 5

**C** 6

**D** 7

**E** 8

**WORKED SOLUTION**

Let  $c$  be the number correct. Then incorrect answers are  $10 - c$ , so  
 $6c - (10 - c) = 32$ . Hence  $7c = 42$  and  $c = 6$ .

**M1-20**

Mock Test 1 - Mathematics 1

Answer **E**

**QUESTION 20**

Emily and Owen are going to a library that is 1 mile from their house. They leave home simultaneously. Emily rides her bicycle to the library at a constant speed of 12 miles per hour. Owen walks to the library at a constant speed of 5 miles per hour. How many minutes before Owen does Emily arrive?

**A** 3

**B** 4

**C** 5

**D** 6

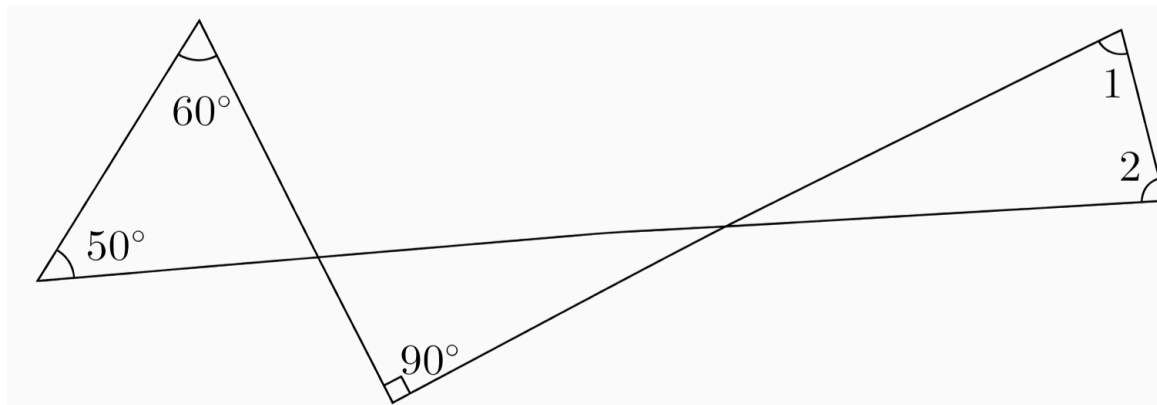
**E** 7

**WORKED SOLUTION**

Emily takes  $1/12$  hour = 5 minutes. Owen takes  $1/5$  hour = 12 minutes. Emily arrives  $12 - 5 = 7$  minutes earlier.

**QUESTION 21**

In the following picture, if  $\angle 1 = \angle 2$ , what is  $\angle 1$ ?



**A**  $60^\circ$

**B**  $65^\circ$

**C**  $70^\circ$

**D**  $75^\circ$

**E**  $80^\circ$

**WORKED SOLUTION**

The left triangle gives the angle at the first intersection as

$180^\circ - 50^\circ - 60^\circ = 70^\circ$ . The next triangle has a right angle, so the angle at the second intersection is  $20^\circ$ . The final small triangle has angles  $x$ ,  $x$ ,  $20^\circ$ , so  $2x = 160^\circ$ , giving  $x = 80^\circ$ .

**QUESTION 22**

Jack, a zoo keeper, is going to give  $N$  peaches to a group of monkeys. If he gives 10 peaches to each monkey, two monkeys will receive no peach. Or, the peaches are exactly enough to give 8 to each monkey. What is the sum of digits of  $N$ ?

**A** 5

**B** 6

**C** 8

**D** 9

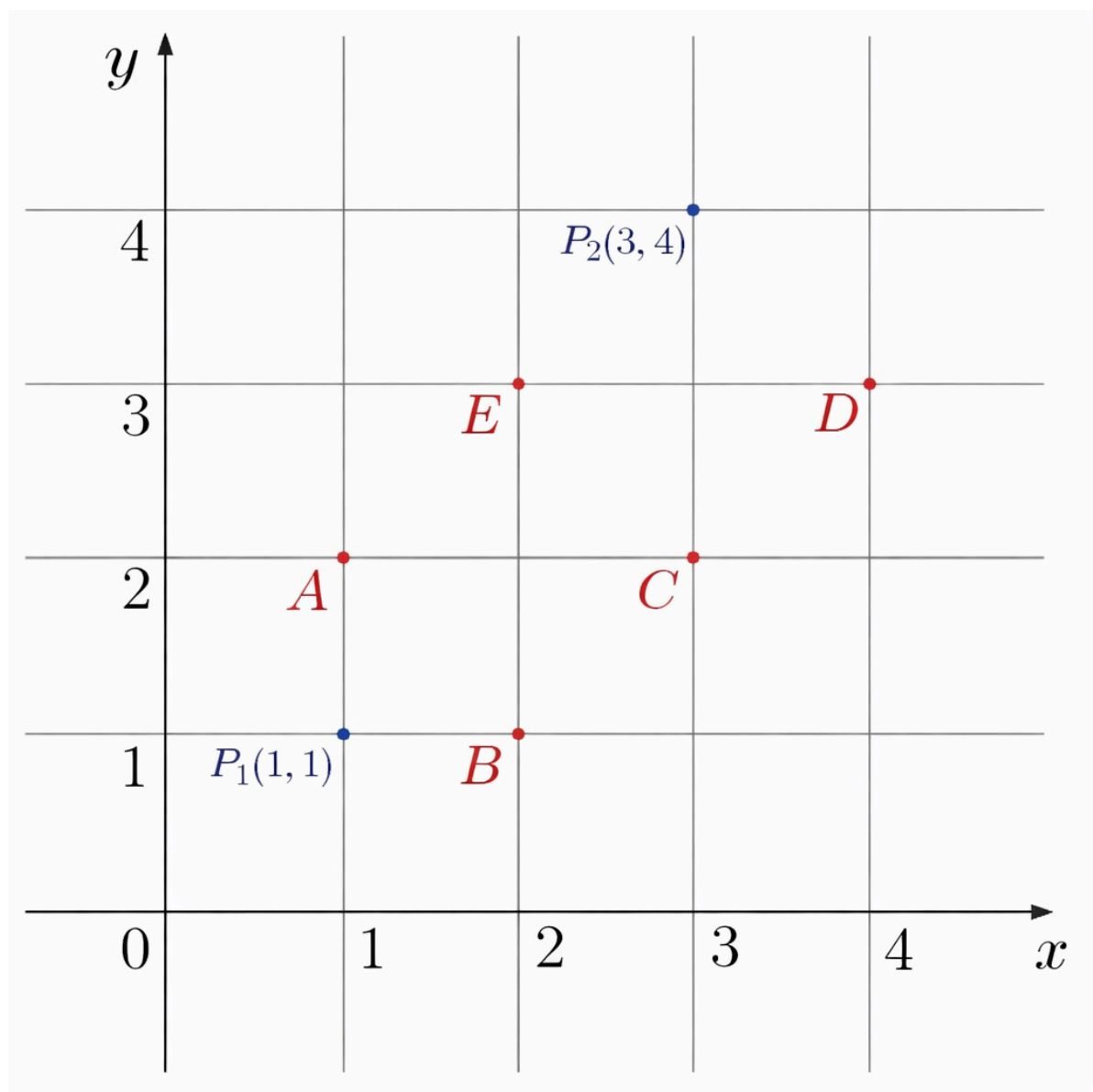
**E** 10

**WORKED SOLUTION**

Let there be  $m$  monkeys. The two descriptions give  $N = 8m$  and  $N = 10(m - 2)$ . Thus  $8m = 10m - 20$ , so  $m = 10$  and  $N = 80$ . Digit sum = 8.

**QUESTION 23**

Every point on the plane can be assigned with a coordinate that has two numbers. For example, the coordinate of point  $P_1$  in the following picture is  $(1, 1)$  and the coordinate of point  $P_2$  is  $(3, 4)$ . Which of the following points has the coordinate  $(2, 3)$ ?



**A** A

**B** B

**C** C

**D** D

**E** E

**WORKED SOLUTION**

Read the grid directly: the point at (2, 3) is labelled E.

**M1-24**

Mock Test 1 - Mathematics 1

Answer **D**

**QUESTION 24**

A pizza-shop offers a basic version of pizza with mozzarella and tomatoes. Two different toppings from the following list must be added: anchovies, artichokes, mushrooms, capers. Moreover, for each pizza three different sizes are available: small, medium, large. How many different types of pizza are available at all?

**A** 11

**B** 12

**C** 15

**D** 18

**E** 36

**WORKED SOLUTION**

Choose 2 toppings from 4:  $\binom{4}{2} = 6$ . For each, there are 3 sizes, so total pizzas  $= 6 \cdot 3 = 18$ .

**M1-25**

Mock Test 1 - Mathematics 1

Answer **E**

**QUESTION 25**

Bag A: 1 Snickers bar + 1 Butterfinger bar.

Bag B: 1 Butterfinger bar + 3 Reese's Peanut Butter Cups.

Bag C: 1 Snickers bar + 2 Reese's Peanut Butter Cups.

If each bag costs 50 cents, what is the price of  
1 Snickers bar + 1 Butterfinger bar + 1 Reese's Peanut Butter Cup,  
in cents?

**A** 40

**B** 45

**C** 50

**D** 55

**E** 60

**WORKED SOLUTION**

Let the prices be  $S$ ,  $B$ ,  $R$ . The bags give  $S + B = 50$ ,  $B + 3R = 50$ , and  $S + 2R = 50$ . Solving gives  $S = 30$ ,  $B = 20$ ,  $R = 10$ , so  $S + B + R = 60$ .

**M1-26**

Mock Test 1 - Mathematics 1

Answer C

**QUESTION 26**

For how many positive integers  $x$  is

$$(x - 1)(x - 4)(x - 9)$$

negative?

**A** 2

**B** 3

**C** 4

**D** 5

**E** 6

**WORKED SOLUTION**

The roots are 1, 4 and 9. Testing the sign in the intervals shows the product is negative for  $x < 1$  and  $4 < x < 9$ . The positive integers satisfying this are 5, 6, 7, 8, so there are 4 values.

**M1-27**

Mock Test 1 - Mathematics 1

Answer C

**QUESTION 27**

A gumball machine contains 9 red, 7 white, and 8 blue gumballs. The least number of gumballs a person must buy to be sure of getting four gumballs of the same color is

**A** 8

**B** 9

**C** 10

**D** 12

**E** 18

**WORKED SOLUTION**

The most gumballs possible without having four of one colour is  $3 + 3 + 3 = 9$ .  
One more guarantees four of some colour, so the least number is 10.

# Physics

27 QUESTIONS · COMPLETE SOLUTIONS

Engineering Mock Test 1 · preserved Physics module. Question wording, option order, answer and solution method are unchanged.

P-01

Mock Test 1 - Physics

Answer **F**

## QUESTION 1

Which of the following statements are **FALSE**?

- A** Electromagnetic waves cause things to heat up.
- B** X-rays and gamma rays can knock electrons out of their orbits.
- C** Loud sounds can make objects vibrate.
- D** Wave power can be used to generate electricity.
- E** Since waves carry energy away, the source of a wave loses energy.
- F** The amplitude of a wave determines its mass.

## WORKED SOLUTION

Waves are not objects with mass. Amplitude affects energy/intensity, not mass, so statement F is false.

**QUESTION 2**

A spacecraft is analysing a newly discovered exoplanet. A rock of unknown mass falls from rest on the planet from a height of 30 m, ignoring air resistance. Given that  $g = 5.4 \text{ m s}^{-2}$  on the planet, calculate the speed of the rock when it hits the ground and the time it took to fall.

|          | Speed ( $\text{ms}^{-1}$ ) | Time (s)   |
|----------|----------------------------|------------|
| <b>A</b> | <b>18</b>                  | <b>3.3</b> |
| B        | 18                         | 3.1        |
| C        | 12                         | 3.3        |
| D        | 10                         | 3.7        |
| E        | 9                          | 2.3        |
| F        | 1                          | 0.3        |

**WORKED SOLUTION**

Use  $v^2 = u^2 + 2as$  with  $u = 0$ :  $v^2 = 2(5.4)(30) = 324$ , so  $v = 18 \text{ m s}^{-1}$ . Also  $s = \frac{1}{2}at^2$ , so  $30 = 2.7t^2$  and  $t \approx 3.3 \text{ s}$ .

**QUESTION 3**

A canoe floating on the sea rises and falls 7 times in 49 seconds. The waves pass it at a speed of  $5 \text{ m s}^{-1}$ . How long are the waves?

**A** 12 m

**B** 22 m

**C** 25 m

**D** 35 m

**E** 57 m

**F** 75 m

**WORKED SOLUTION**

The period is  $T = 49/7 = 7$  s. Since  $v = f\lambda = \lambda/T$ ,  $\lambda = vT = 5 \cdot 7 = 35$  m.

**P-04**

Mock Test 1 - Physics

Answer **E**

**QUESTION 4**

Miss Orrell lifts her 37.5 kg bike for a distance of 1.3 m in 5 s. The acceleration of free fall is  $10 \text{ m s}^{-2}$ . What is the average power that she develops?

**A** 9.8 W

**B** 12.9 W

**C** 57.9 W

**D** 79.5 W

**E** 97.5 W

**F** 98.0 W

**WORKED SOLUTION**

Work done =  $mgh = 37.5 \cdot 10 \cdot 1.3 = 487.5 \text{ J}$ . Power =  $487.5 / 5 = 97.5 \text{ W}$ .

**P-05**

Mock Test 1 - Physics

Answer **G**

**QUESTION 5**

A truck accelerates at  $5.6 \text{ m s}^{-2}$  from rest for 8 seconds. Calculate the final speed and the distance travelled in 8 seconds.

|          | Final Speed ( $\text{ms}^{-1}$ ) | Distance (m) |
|----------|----------------------------------|--------------|
| <b>A</b> | 40.8                             | 119.2        |
| <b>B</b> | 40.8                             | 129.6        |
| <b>C</b> | 42.8                             | 179.2        |
| <b>D</b> | 44.1                             | 139.2        |
| <b>E</b> | 44.1                             | 179.7        |
| <b>F</b> | 44.2                             | 129.2        |
| <b>G</b> | <b>44.8</b>                      | <b>179.2</b> |
| <b>H</b> | 44.8                             | 179.7        |

**WORKED SOLUTION**

From rest,  $v = at = 5.6 \cdot 8 = 44.8 \text{ m s}^{-1}$ . Distance

$$s = \frac{1}{2}at^2 = \frac{1}{2} \cdot 5.6 \cdot 64 = 179.2 \text{ m}.$$

**QUESTION 6**

Which of the following statements is true when a sky diver jumps out of a plane?

- A** The sky diver leaves the plane and will accelerate until the air resistance is greater than their weight.
- B** The sky diver leaves the plane and will accelerate until the air resistance is less than their weight.
- C** The sky diver leaves the plane and will accelerate until the air resistance equals their weight.
- D** The sky diver leaves the plane and will accelerate until the air resistance equals their weight squared.
- E** The sky diver will travel at a constant velocity after leaving the plane.

**WORKED SOLUTION**

Initially weight is greater than air resistance, so the skydiver accelerates. As speed rises, air resistance increases until it equals weight; then acceleration becomes zero.

**QUESTION 7**

A 100 g apple falls from rest on Isaac's head from a height of 20 m, ignoring air resistance. Calculate the apple's momentum before the point of impact. Take  $g = 10 \text{ m s}^{-2}$ .

**A**  $0.1 \text{ kg m s}^{-1}$

**B**  $0.2 \text{ kg m s}^{-1}$

**C**  $1 \text{ kg m s}^{-1}$

**D**  $2 \text{ kg m s}^{-1}$

**E**  $10 \text{ kg m s}^{-1}$

**F**  $20 \text{ kg m s}^{-1}$

#### WORKED SOLUTION

The apple mass is  $0.1 \text{ kg}$ . Its speed before impact is

$$v = \sqrt{2gh} = \sqrt{2 \cdot 10 \cdot 20} = 20 \text{ m s}^{-1}. \text{ Momentum}$$

$$p = mv = 0.1 \cdot 20 = 2 \text{ kg m s}^{-1}.$$

**P-08**

Mock Test 1 - Physics

Answer E

#### QUESTION 8

Which of the following do all electromagnetic waves have in common?

1. They can travel through a vacuum.
2. They can be reflected.
3. They are the same length.
4. They have the same amount of energy.
5. They can be polarised.

**A** 1, 2 and 3 only

**B** 1, 2, 3 and 4 only

**C** 4 and 5 only

**D** 3 and 4 only

**E** 1, 2 and 5 only

**F** 1 and 5 only

#### WORKED SOLUTION

All electromagnetic waves can travel through vacuum, reflect, and be polarised. They do not all have the same wavelength or energy.

**P-09**

Mock Test 1 - Physics

Answer **D**

#### QUESTION 9

A battery with an internal resistance of  $0.8\ \Omega$  and e.m.f of  $36\ \text{V}$  is used to power a drill with resistance  $1\ \Omega$ . What is the current in the circuit when the drill is connected to the power supply?

**A** 5 A

**B** 10 A

**C** 15 A

**D** 20 A

**E** 25 A

**F** 30 A

#### WORKED SOLUTION

The internal and drill resistances are in series, so  $R_{\text{total}} = 0.8 + 1 = 1.8 \Omega$ .

Current  $I = V/R = 36/1.8 = 20 \text{ A}$ .

**P-10**

Mock Test 1 - Physics

Answer D

#### QUESTION 10

Officer Bailey throws a 20 g dart at a speed of  $100 \text{ m s}^{-1}$ . It strikes the dartboard and is brought to rest in 10 milliseconds. Calculate the average force exerted on the dart by the dartboard.

**A** 0.2 N

**B** 2 N

**C** 20 N

**D** 200 N

**E** 2000 N

**F** 20000 N

**WORKED SOLUTION**

Mass = 0.020 kg, change in velocity =  $100 \text{ m s}^{-1}$ , and time = 0.010 s.

Average force  $F = \Delta p / \Delta t = 0.020 \cdot 100 / 0.010 = 200 \text{ N}$ .

**P-11**

Mock Test 1 - Physics

Answer **F**

**QUESTION 11**

Professor Huang lifts a 50 kg bag through a distance of 0.7 m in 3 s. What average power does she develop to 3 significant figures? Take  $g = 10 \text{ m s}^{-2}$ .

**A** 112 W

**B** 113 W

**C** 114 W

**D** 115 W

**E** 116 W

**F** 117 W

**WORKED SOLUTION**

Work done =  $mgh = 50 \cdot 10 \cdot 0.7 = 350 \text{ J}$ . Power =  $350/3 = 116.7 \text{ W}$ , which is  $117 \text{ W}$  to 3 significant figures.

**P-12**

Mock Test 1 - Physics

Answer **B**

**QUESTION 12**

An electric scooter is travelling at a speed of  $30 \text{ m s}^{-1}$ . The motor provides a driving force of  $300 \text{ N}$  in the direction of motion. Assuming all electrical power is converted into mechanical power, and given that the motor runs at  $200 \text{ V}$ , calculate the current in the scooter.

**A** 4.5 A

**B** 45 A

**C** 450 A

**D** 4500 A

**E** 45000 A

**F** More information needed.

**WORKED SOLUTION**

Use the motor driving force for power:  $P = Fv = 300 \cdot 30 = 9000 \text{ W}$ . Since  $P = VI$ ,  $I = 9000/200 = 45 \text{ A}$ .

**QUESTION 13**

Which of the following statements about the physical definition of work are correct?

1. Work done =  $\frac{\text{Force}}{\text{distance}}$
2. The unit of work is equivalent to  $\text{kg m s}^{-2}$ .
3. Work is defined as a force causing displacement of the body upon which it acts.

**A** Only 1

**B** Only 2

**C** Only 3

**D** 1 and 2

**E** 2 and 3

**F** 1 and 3

**WORKED SOLUTION**

Work is force times displacement in the direction of the force, not force divided by distance. Its unit is the joule,  $\text{kg m}^2\text{s}^{-2}$ , not  $\text{kg m s}^{-2}$ . Only statement 3 is correct.

**QUESTION 14**

Which of the following statements about kinetic energy are correct?

1. It is defined as  $E_k = \frac{mv^2}{2}$ .
2. The unit of kinetic energy is equivalent to  $\text{Pa} \times \text{m}^3$ .
3. Kinetic energy is equal to the amount of energy needed to decelerate the body in question from its current speed.

**A** Only 1

**B** Only 2

**C** Only 3

**D** 1 and 2

**E** 2 and 3

**F** 1 and 3

**G** 1, 2 and 3

#### WORKED SOLUTION

$E_k = \frac{1}{2}mv^2$ . Also  $\text{Pa} \cdot \text{m}^3 = (\text{N}/\text{m}^2)\text{m}^3 = \text{N m} = \text{J}$ . Kinetic energy is the work needed to bring the body to rest, so all three are correct.

**P-15**

Mock Test 1 - Physics

Answer D

#### QUESTION 15

In relation to radiation, which of the following statements is **FALSE**?

**A** Radiation is the emission of energy in the form of waves or particles.

**B** Radiation can be either ionizing or non-ionizing.

**C** Gamma radiation has very high energy.

**D** Alpha radiation is of higher energy than beta radiation.

**E** X-rays are an example of wave radiation.

#### WORKED SOLUTION

Radiation can be waves or particles, and gamma rays and X-rays are high-energy electromagnetic radiation. The statement that alpha radiation is always of higher energy than beta radiation is not a reliable general definition, so D is false.

**P-16**

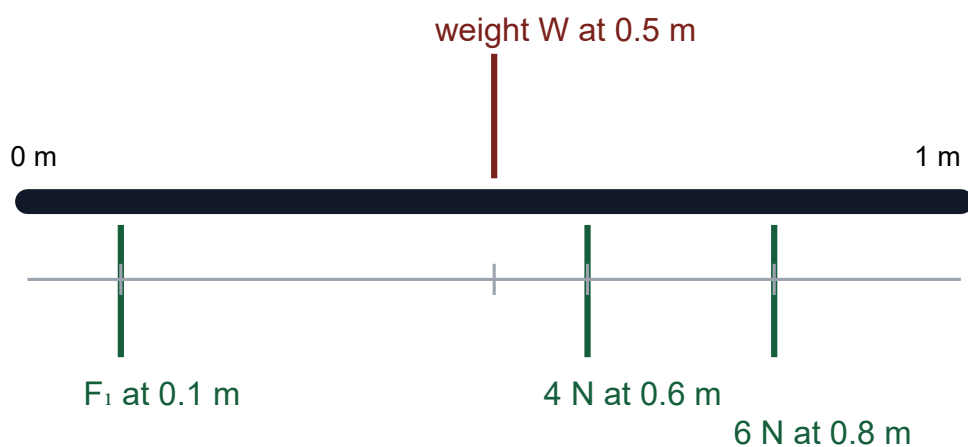
Mock Test 1 - Physics

Answer E

#### QUESTION 16

A uniform rod of length 1 m is balanced on 3 supports so that it is in equilibrium. The position of the first support is at  $x_1 = 0.1$  m, where  $x$  is the distance along the rod. The second support is at  $x_2 = 0.6$  m and applies a force,  $F_2$ , of 4 N to the rod. The third support is at  $x_3 = 0.8$  m and applies a force,  $F_3$ , of 6 N to the rod.

What is the force applied by the first support,  $F_1$ ?



**A** 4.5 N

**B** 5.0 N

**C** 5.3 N

**D** 6.7 N

**E** 5.5 N

**F** 11.0 N

**G** 10.5 N

**H** 7.3 N

#### WORKED SOLUTION

Let the rod weight be  $W$ . Force balance gives  $W = F_1 + 4 + 6 = F_1 + 10$ .

Taking moments about the left end:  $0.1F_1 + 0.6(4) + 0.8(6) = 0.5W$ . Hence  $0.1F_1 + 7.2 = 0.5(F_1 + 10)$ , so  $F_1 = 5.5$  N.

**P-17**

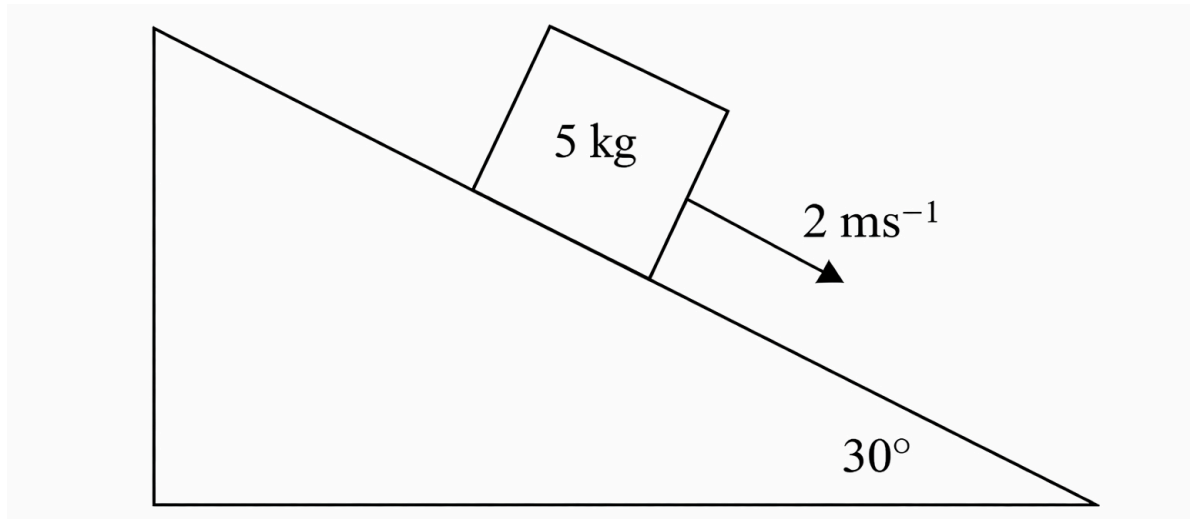
Mock Test 1 - Physics

Answer **B**

#### QUESTION 17

A block of mass 5 kg slides down a rough slope angled at  $30^\circ$  to the horizontal. The block moves at a constant velocity of  $2 \text{ m s}^{-1}$ . (Take  $g = 10 \text{ m s}^{-2}$ .)

What is the frictional force on the block?



**A** 14 N

**B** 25 N

**C**  $\frac{50}{\sqrt{2}} \text{ N}$

**D**  $50\sqrt{2} \text{ N}$

**E**  $25\sqrt{3} \text{ N}$

**F** 55 N

**G**  $50\sqrt{3} \text{ N}$

**H** 50 N

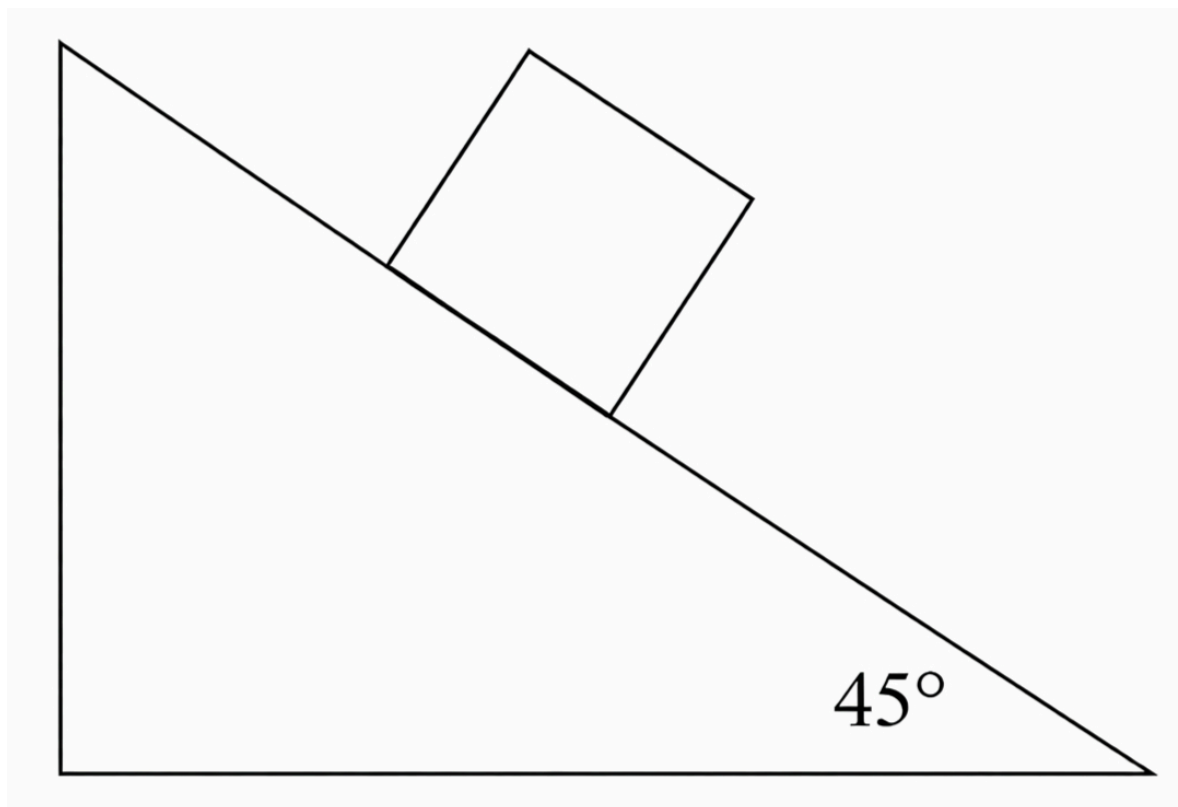
#### WORKED SOLUTION

Constant velocity means zero resultant force along the slope. The downslope component of weight is  $mg \sin 30^\circ = 5 \cdot 10 \cdot \frac{1}{2} = 25 \text{ N}$ , so friction is 25 N upslope.

**QUESTION 18**

A block slides down a smooth slope angled at  $45^\circ$  to the horizontal. The block is initially stationary. (Take  $g = 10 \text{ m s}^{-2}$ .)

What is the velocity of the block after 10 s?



**A**  $50 \text{ m s}^{-1}$

**B**  $40 \text{ m s}^{-1}$

**C**  $\frac{100}{\sqrt{2}} \text{ m s}^{-1}$

**D**  $\frac{200}{\sqrt{3}} \text{ m s}^{-1}$

**E**  $\frac{100}{\sqrt{3}} \text{ m s}^{-1}$

**F**  $75 \text{ m s}^{-1}$

**G**  $\frac{200}{3} \text{ m s}^{-1}$

**H**  $\frac{200}{\sqrt{2}} \text{ m s}^{-1}$

**WORKED SOLUTION**

On a smooth slope, acceleration down the plane is  $g \sin 45^\circ = 10/\sqrt{2}$ . Starting from rest for 10 seconds gives  $v = at = 100/\sqrt{2} \text{ m s}^{-1}$ .

**P-19**

Mock Test 1 - Physics

Answer **G**

**QUESTION 19**

An object of mass  $10 \text{ kg}$  falls from a great height. If the force due to air resistance can be modelled as  $F_a = 0.5u \text{ N}$  where  $u$  is the velocity of the free-falling object, what is the terminal velocity of the object? (Take  $g = 10 \text{ N kg}^{-1}$ .)

**A**  $700 \text{ m s}^{-1}$

**B**  $150 \text{ m s}^{-1}$

**C**  $50 \text{ m s}^{-1}$

**D**  $100 \text{ m s}^{-1}$

**E**  $800 \text{ m s}^{-1}$

**F**  $400 \text{ m s}^{-1}$

**G**  $200 \text{ m s}^{-1}$

**H**  $500 \text{ m s}^{-1}$

**WORKED SOLUTION**

At terminal velocity, air resistance equals weight:  $0.5u = mg = 10 \cdot 10 = 100$ .  
Therefore  $u = 200 \text{ m s}^{-1}$ .

**P-20**

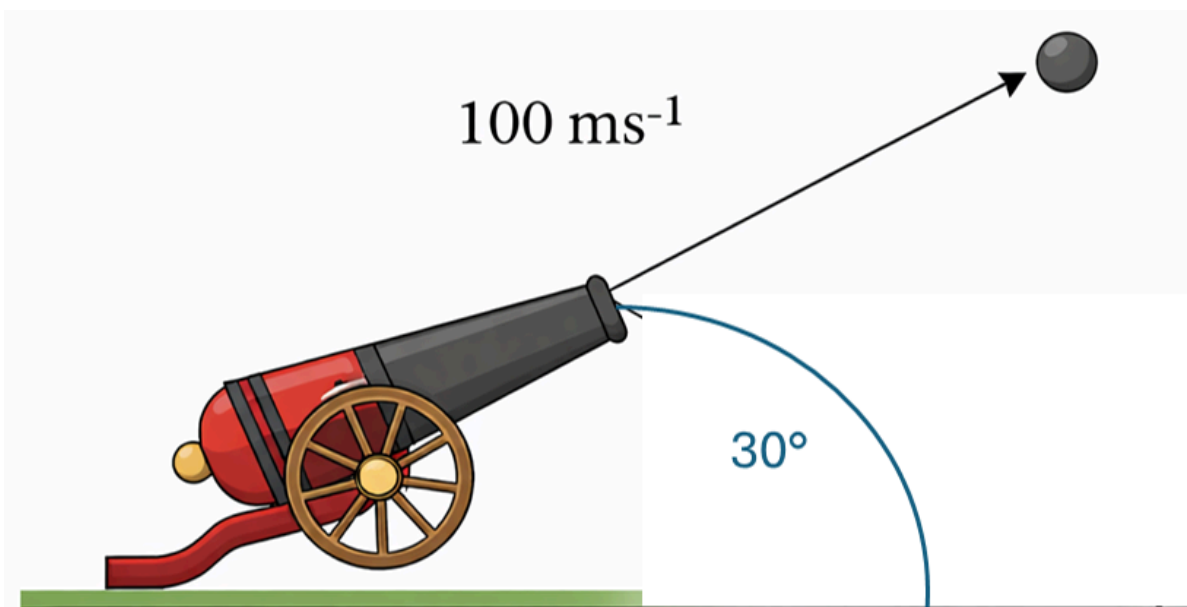
Mock Test 1 - Physics

Answer **D**

**QUESTION 20**

A cannonball is fired from ground level at  $30^\circ$  to the horizontal with an initial velocity of  $100 \text{ m s}^{-1}$ . The ground is flat and air resistance is negligible. (Take  $g = 10 \text{ N kg}^{-1}$ .)

How far does the cannonball travel horizontally before it hits the ground?



**A**  $250 \text{ m}$

**B**  $250\sqrt{2} \text{ m}$

**C** 300 m

**D**  $500\sqrt{3}$  m

**E** 500 m

**F**  $\frac{500}{\sqrt{2}}$  m

**G**  $600\sqrt{3}$  m

**H**  $600\sqrt{2}$  m

#### WORKED SOLUTION

Use projectile range  $R = \frac{u^2 \sin 2\theta}{g}$ . Thus

$$R = \frac{100^2 \sin 60^\circ}{10} = 1000 \cdot \frac{\sqrt{3}}{2} = 500\sqrt{3} \text{ m.}$$

**P-21**

Mock Test 1 - Physics

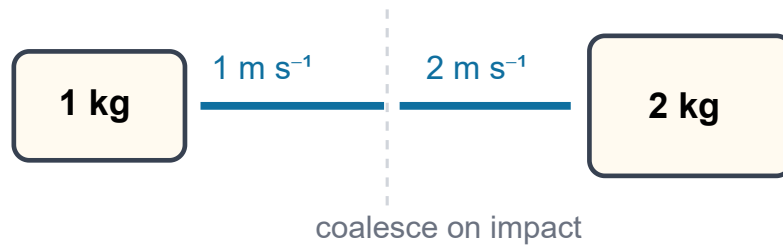
Answer E

#### QUESTION 21

Two blocks, one of mass 1 kg and one of mass 2 kg, move directly towards each other and collide inelastically. Before collision, their velocities have magnitudes of  $1 \text{ m s}^{-1}$  and  $2 \text{ m s}^{-1}$  respectively.

Assuming the two blocks coalesce on impact, what fraction of the kinetic energy is lost (e.g. to heat)?

### Before collision



**A** 1

**B** 0

**C**  $\frac{1}{3}$

**D**  $\frac{1}{2}$

**E**  $\frac{2}{3}$

**F**  $\frac{1}{4}$

**G**  $\frac{3}{4}$

**H**  $\frac{2}{5}$

#### WORKED SOLUTION

Initial momentum is  $1(1) - 2(2) = -3$ , so after coalescing,  
 $v = -3/(1 + 2) = -1 \text{ m s}^{-1}$ . Initial KE is  $\frac{1}{2}(1)(1^2) + \frac{1}{2}(2)(2^2) = 4.5$ . Final  
KE is  $\frac{1}{2}(3)(1^2) = 1.5$ . Fraction lost =  $(4.5 - 1.5)/4.5 = 2/3$ .

**QUESTION 22**

An object of mass 20 kg falls 100 m and experiences an average resistive force of 40 N due to air resistance. If the object is initially at rest, what is its final velocity? (Take  $g = 10 \text{ N kg}^{-1}$ .)

**A**  $60 \text{ m s}^{-1}$

**B**  $420 \text{ m s}^{-1}$

**C**  $360 \text{ m s}^{-1}$

**D**  $200 \text{ m s}^{-1}$

**E**  $80 \text{ m s}^{-1}$

**F**  $500 \text{ m s}^{-1}$

**G**  $40 \text{ m s}^{-1}$

**H**  $800 \text{ m s}^{-1}$

**WORKED SOLUTION**

Energy lost to air resistance is  $40 \cdot 100 = 4000 \text{ J}$ . Gravitational energy lost is  $mgh = 20 \cdot 10 \cdot 100 = 20000 \text{ J}$ . Final KE =  $16000 \text{ J} = \frac{1}{2}(20)v^2$ , so  $v = 40 \text{ m s}^{-1}$ .

**QUESTION 23**

A velocity profile is given by

$$v = \frac{(5t + 3)^2}{t^{\frac{1}{2}}}$$

(SI units). Find the total distance travelled from  $t = 1$  to  $t = 4$  s.

**A** 490 m

**B** 955 m

**C** 1082 m

**D** 683 m

**E** 575 m

**F** 23 m

**G** 1820 m

**H** 468 m

**WORKED SOLUTION**

Rewrite  $v = (5t + 3)^2 t^{-1/2} = 25t^{3/2} + 30t^{1/2} + 9t^{-1/2}$ . Integrating gives  $10t^{5/2} + 20t^{3/2} + 18t^{1/2}$ . From 1 to 4:  
 $(320 + 160 + 36) - (10 + 20 + 18) = 468$  m.

**QUESTION 24**

A spring is used to lift  $20 \text{ cm}^3$  of loosely packed sand in a small bucket. The spring constant,  $k$ , is  $4 \text{ N m}^{-1}$  and the sand causes the spring to extend by  $10 \text{ cm}$ . What is the density of the sand at this packing fraction? (Take  $g = 10 \text{ N kg}^{-1}$ .)

**A**  $3200 \text{ kg m}^{-3}$

**B**  $32000 \text{ kg m}^{-3}$

**C**  $2000 \text{ kg m}^{-3}$

**D**  $1000 \text{ kg m}^{-3}$

**E**  $20000 \text{ kg m}^{-3}$

**F**  $2500 \text{ kg m}^{-3}$

**G**  $4500 \text{ kg m}^{-3}$

**H**  $800 \text{ kg m}^{-3}$

**WORKED SOLUTION**

The spring force is  $F = kx = 4 \cdot 0.10 = 0.4 \text{ N}$ , so mass  $m = F / g = 0.04 \text{ kg}$ .

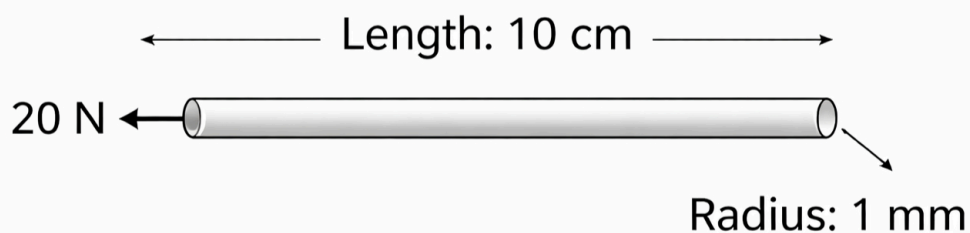
Volume  $20 \text{ cm}^3 = 2.0 \times 10^{-5} \text{ m}^3$ . Density

$= 0.04 / (2.0 \times 10^{-5}) = 2000 \text{ kg m}^{-3}$ .

**QUESTION 25**

A cylindrical wire of radius 1 mm, and length 10 cm is extended by applying a tension of 20 N to it, resulting in an elastic extension of 2 mm.

What is the Young's modulus of the material?



**A**  $2\pi \text{ GPa}$

**B**  $\frac{2}{\pi} \text{ GPa}$

**C**  $\pi \text{ GPa}$

**D**  $\frac{1}{\pi} \text{ GPa}$

**E**  $20\pi \text{ MPa}$

**F**  $\frac{20}{\pi} \text{ MPa}$

**G**  $10\pi \text{ GPa}$

**H**  $\frac{10}{\pi} \text{ GPa}$

**WORKED SOLUTION**

Young's modulus  $E = \frac{FL}{A\Delta L}$ . With  $F = 20$ ,  $L = 0.10$ ,  $A = \pi(10^{-3})^2$ , and  $\Delta L = 2 \times 10^{-3}$ ,  $E = \frac{1}{\pi} \times 10^9 \text{ Pa} = \frac{1}{\pi} \text{ GPa}$ .

**P-26**

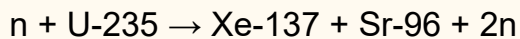
Mock Test 1 - Physics

Answer C

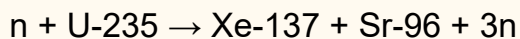
**QUESTION 26**

A uranium-235 nucleus can undergo fission to produce two smaller nuclei. Which of the diagrams, if any, could represent this process?

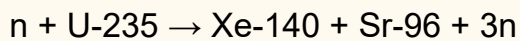
**Diagram 1**



**Diagram 2**



**Diagram 3**



**A** none of them

**B** 1 only

**C** 2 only

**D** 3 only

**E** 1 and 2 only

**F** 1 and 3 only

#### WORKED SOLUTION

Check conservation of nucleon number and proton number. Diagram 2 conserves both:  $1 + 235 = 137 + 96 + 3$  and  $92 = 54 + 38$ . Diagrams 1 and 3 do not conserve mass number as drawn.

**P-27**

Mock Test 1 - Physics

Answer **B**

#### QUESTION 27

A wave travels at 30 m in 10 seconds and has a frequency,  $f$ , of 2 Hz. How many complete wave cycles occur if the wave travels 120 m?

**A** 100

**B** 80

**C** 50

**D** 40

**E** 120

**F** 20

**G** 30

**H** 240

#### WORKED SOLUTION

The wave speed is  $30/10 = 3 \text{ m s}^{-1}$ . Wavelength  $\lambda = v/f = 3/2 = 1.5 \text{ m}$ .  
Over 120 m, the number of cycles is  $120/1.5 = 80$ .

# Biology

## 27 QUESTIONS • COMPLETE SOLUTIONS

Preserved Biology Easy source set. Question wording, option order, answer and solution method are unchanged.

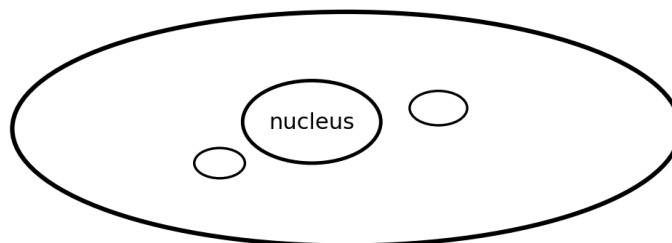
**B-01**

Microscopy

Answer C

**QUESTION 1**

The diagram shows a cell viewed using a microscope. The scale bar represents **10  $\mu\text{m}$** .

**Cell image**10  $\mu\text{m}$ 

Approximately how long is the cell along its longest axis?

**A** 20  $\mu\text{m}$ **B** 40  $\mu\text{m}$

**C** 60  $\mu\text{m}$

**D** 80  $\mu\text{m}$

**E** 100  $\mu\text{m}$

#### WORKED SOLUTION

The cell is approximately 5.4 scale-bar lengths long.

$$5.4 \times 10 \mu\text{m} \approx 54 \mu\text{m}$$

The nearest option is **60  $\mu\text{m}$** .

**B-02**

Levels of organisation

Answer **A**

#### QUESTION 2

Which sequence places the structures in order from **smallest to largest**?

1. circulatory system
2. heart
3. mitochondrion
4. muscle cell
5. muscle tissue

**A** 3, 4, 5, 2, 1

**B** 4, 3, 5, 2, 1

**C** 3, 5, 4, 2, 1

**D** 3, 4, 2, 5, 1

**E** 4, 5, 3, 2, 1

#### WORKED SOLUTION

The hierarchy is:

sub-cellular component < cell < tissue < organ < organ system

Therefore the correct order is **mitochondrion** → **muscle cell** → **muscle tissue** → **heart** → **circulatory system**.

**B-03**

Gas exchange

Answer **H**

#### QUESTION 3

Which features increase the rate at which oxygen moves from an alveolus into the blood?

1. A large alveolar surface area.
2. A very thin barrier between alveolar air and blood.
3. Continuous blood flow carrying oxygenated blood away from the alveolus.

**A** none

**B** 1 only

**C** 2 only

**D** 3 only

**E** 1 and 2 only

**F** 1 and 3 only

**G** 2 and 3 only

**H** 1, 2 and 3

#### WORKED SOLUTION

All three increase diffusion rate. A large surface area provides more area for exchange, a thin barrier shortens diffusion distance, and blood flow removes oxygen that has entered the blood, helping maintain a steep concentration gradient.

**B-04**

Active transport

Answer C

#### QUESTION 4

Glucose is being absorbed from the small intestine.

[glucose in intestinal lumen] < [glucose inside an epithelial cell]

but glucose continues to enter the epithelial cell.

Which row best describes this movement and predicts the effect of strongly inhibiting respiration in the epithelial cell?

**A** diffusion; uptake increases

**B** osmosis; uptake decreases

**C** active transport; uptake decreases

**D** active transport; no effect

**E** diffusion; no effect

#### WORKED SOLUTION

Glucose is moving from lower concentration outside the cell to higher concentration inside the cell, so it is moving against its concentration gradient. This requires **active transport**.

Active transport requires energy from respiration, so strongly inhibiting respiration should **decrease uptake**.

**B-05**

Mitosis and meiosis

Answer E

### QUESTION 5

A diploid cell contains **8 chromosomes**.

Which row correctly describes the products of mitosis and meiosis?

- A** mitosis: two cells, 4 chromosomes each; meiosis: four cells, 8 chromosomes each
- B** mitosis: four cells, 8 chromosomes each; meiosis: two cells, 4 chromosomes each
- C** mitosis: two cells, 8 chromosomes each; meiosis: two cells, 4 chromosomes each
- D** mitosis: two cells, 4 chromosomes each; meiosis: four cells, 4 chromosomes each
- E** mitosis: two cells, 8 chromosomes each; meiosis: four cells, 4 chromosomes each

### WORKED SOLUTION

Mitosis produces two daughter cells that retain the diploid chromosome number, so each has **8 chromosomes**.

Meiosis halves the chromosome number and produces four cells, so each has **4 chromosomes**.

**B-06**

Sex determination

Answer E

**QUESTION 6**

Assume that the probability of a child being genetically male or female is equal and that the sex of one child does not affect the next.

What is the probability that, in a family's **next three children**, at least one is female?

**A** 1/8**B** 1/4**C** 1/2**D** 3/4**E** 7/8**WORKED SOLUTION**

It is easiest to calculate the opposite event: all three children are male.

$$P(\text{all male}) = (1/2)^3 = 1/8$$

$$P(\text{at least one female}) = 1 - 1/8 = 7/8$$

**B-07**

Inheritance

Answer C

**QUESTION 7**

In a plant species, tall stems are controlled by a dominant allele **T**, while short stems are controlled by the recessive allele **t**.

Two tall plants produce a short offspring.

What is the probability that their next offspring will also be short?

**A** 0

**B** 1/8

**C** 1/4

**D** 1/2

**E** 3/4

#### WORKED SOLUTION

A short plant must have genotype **tt**. Since two tall plants produced a short offspring, both tall parents must be heterozygous:

$$Tt \times Tt$$

The possible genotypes are TT, Tt, Tt and tt, so:

$$P(tt) = 1/4$$

**B-08**

DNA base composition

Answer D

#### QUESTION 8

A double-stranded DNA molecule contains **200 base pairs**.

Cytosine accounts for **22%** of all bases.

How many thymine bases are present?

**A** 44

**B** 56

**C** 88

**D** 112

**E** 156

#### WORKED SOLUTION

There are 400 bases in total. In double-stranded DNA, C and G occur in equal proportions, so:

$$C = 22\%, \quad G = 22\%$$

Thus  $C + G = 44\%$ , leaving  $56\%$  for  $A + T$ . Therefore  $T = 28\%$ .

$$0.28 \times 400 = \mathbf{112}$$

**B-09**

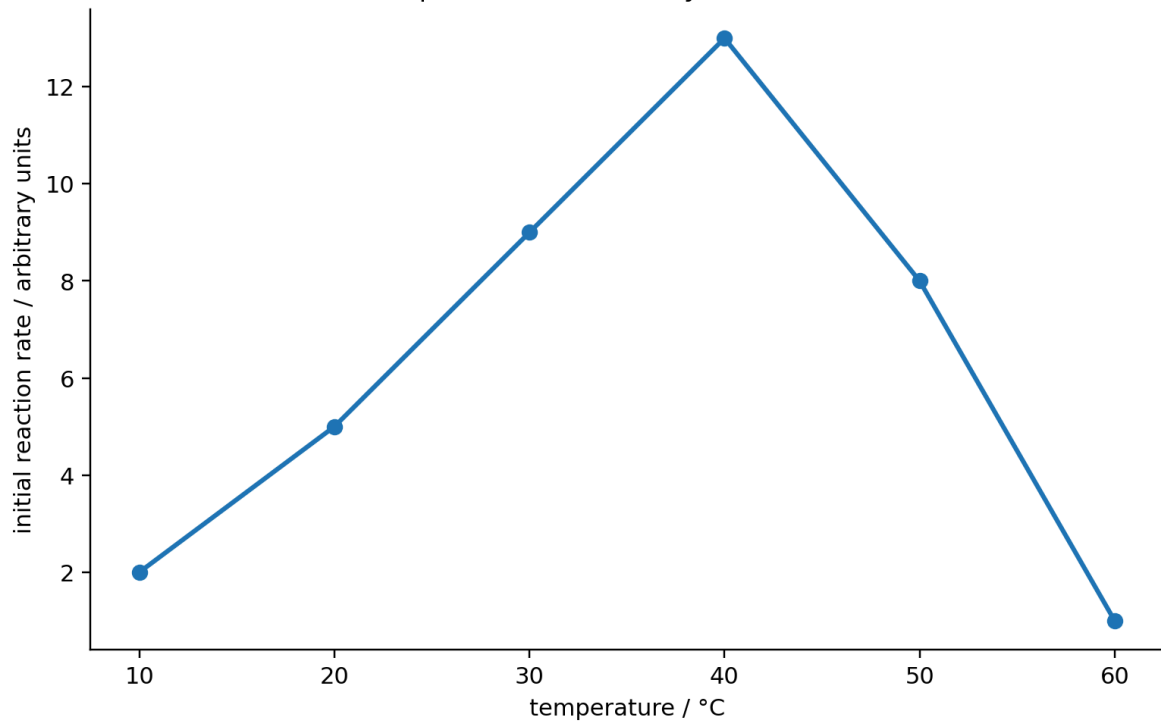
Enzymes

Answer **B**

#### QUESTION 9

The graph shows the initial rate of an enzyme-controlled reaction at six temperatures.

Effect of temperature on an enzyme-controlled reaction



Which statements are justified by these data?

1. The measured reaction rate increases between 10°C and 40°C.
2. The exact optimum temperature of the enzyme must be 40°C.
3. At 60°C, every enzyme molecule must have been permanently denatured.

**A** none

**B** 1 only

**C** 2 only

**D** 3 only

**E** 1 and 2 only

**F** 1 and 3 only

**G** 2 and 3 only

**H** 1, 2 and 3

#### WORKED SOLUTION

Statement 1 is directly supported: the measured rate rises between 10°C and 40°C.

Statement 2 is too strong because temperatures between the measured values were not tested; the true optimum could be slightly below or above 40°C.

Statement 3 is also too strong. The graph still shows a measurable reaction at 60°C, and in any case it does not establish that every enzyme molecule has been permanently denatured.

Therefore **1 only**.

**B-10**

Digestion

Answer B

#### QUESTION 10

Which row correctly identifies an enzyme, its substrate and one of its products?

**A** amylase; protein; amino acids

**B** protease; protein; amino acids

**C** lipase; starch; simple sugars

**D** protease; lipid; fatty acids

**E** lipase; protein; glycerol

#### WORKED SOLUTION

Proteases catalyse the breakdown of proteins into amino acids. Therefore the correct combination is **protease; protein; amino acids**.

**B-11**

Contraception

Answer **E**

**QUESTION 11**

Which statements are correct?

1. Hormonal contraception containing oestrogen and progesterone can inhibit FSH production and help prevent ovulation.
2. Condoms are a non-hormonal form of contraception and can reduce transmission of sexually transmitted infections.
3. Hormonal contraceptives work mainly by directly killing sperm cells.

**A** none

**B** 1 only

**C** 2 only

**D** 3 only

**E** 1 and 2 only

**F** 1 and 3 only

**G** 2 and 3 only

**H** 1, 2 and 3

**WORKED SOLUTION**

Statements 1 and 2 are correct. Hormonal contraception can suppress FSH and therefore help prevent ovulation. Condoms are a barrier method and can also reduce transmission of many sexually transmitted infections.

Hormonal contraception does not principally work by directly killing sperm cells.

**B-12**

Ciliated cells

Answer **G**

**QUESTION 12**

Long-term smoking can damage the cilia lining parts of the respiratory tract.

Which consequences could result directly from reduced ciliary activity?

1. The percentage of oxygen in inhaled air decreases.
2. Mucus may accumulate in the airways.
3. Microorganisms trapped in mucus may be removed less effectively.

**A** none

**B** 1 only

**C** 2 only

**D** 3 only

**E** 1 and 2 only

**F** 1 and 3 only

**G** 2 and 3 only

**H** 1, 2 and 3

**WORKED SOLUTION**

Damaged cilia do not change the percentage of oxygen in the external air being inhaled. However, mucus is cleared less effectively, so it can accumulate and microorganisms trapped in it may remain in the respiratory tract for longer.

Therefore **2 and 3 only**.

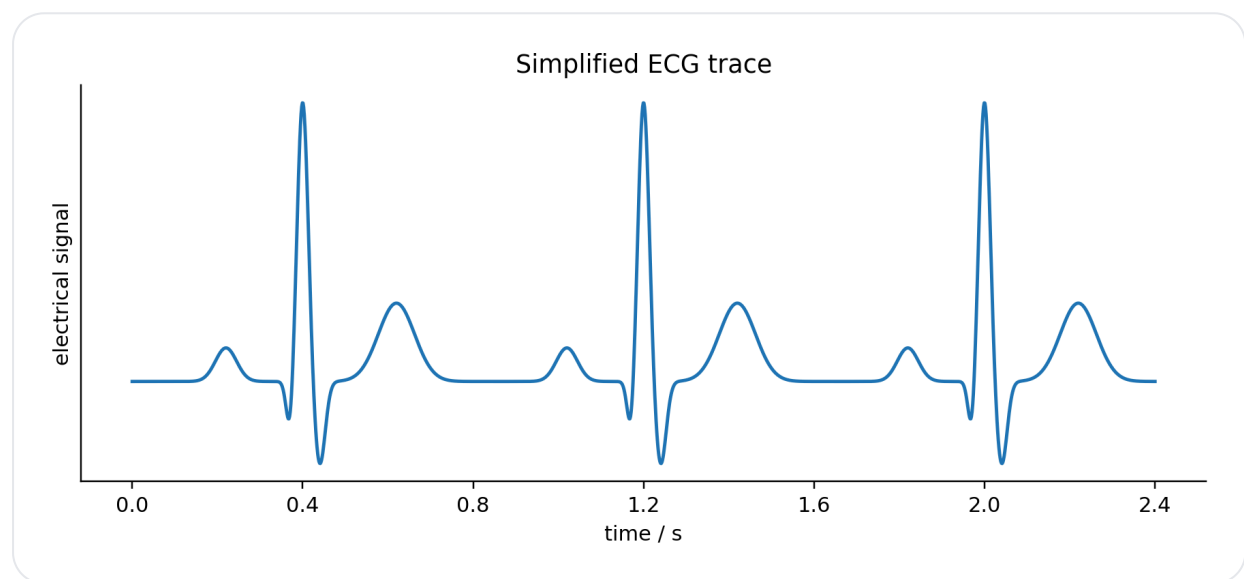
**B-13**

ECG and heart rate

Answer C

### QUESTION 13

The trace shows a simplified ECG.



The large R-peaks are approximately **0.80 s** apart.

What is the heart rate?

**A** 48 beats  $\text{min}^{-1}$

**B** 60 beats  $\text{min}^{-1}$

**C** 75 beats  $\text{min}^{-1}$

**D** 80 beats  $\text{min}^{-1}$

**E** 96 beats  $\text{min}^{-1}$

#### WORKED SOLUTION

One heartbeat takes approximately 0.80 s.

$$\text{heart rate} = 60 / 0.80 = \mathbf{75 \text{ beats min}^{-1}}$$

**B-14**

Blood

Answer **D**

#### QUESTION 14

A patient has:

- a much lower than normal platelet count;
- a much lower than normal red blood cell count.

Which combination of effects is most likely?

**A** blood clotting faster; oxygen-carrying capacity increased

**B** blood clotting faster; oxygen-carrying capacity decreased

**C** blood clotting slower; oxygen-carrying capacity unchanged

**D** blood clotting slower; oxygen-carrying capacity decreased

**E** blood clotting unchanged; oxygen-carrying capacity decreased

#### WORKED SOLUTION

Platelets are required for normal blood clotting, so a low platelet count tends to make clotting slower or less effective.

Red blood cells contain haemoglobin and carry oxygen, so a reduced red blood cell count lowers the blood's oxygen-carrying capacity.

**B-15**

Respiration

Answer **B**

#### QUESTION 15

Four flasks are prepared.

| flask | contents                        | temperature |
|-------|---------------------------------|-------------|
| A     | living yeast + glucose solution | 20°C        |
| B     | living yeast + glucose solution | 35°C        |
| C     | boiled yeast + glucose solution | 35°C        |
| D     | living yeast + water only       | 35°C        |

Assume 35°C is closer to the optimum temperature for the enzymes involved in yeast respiration than 20°C, and neither temperature causes denaturation.

Which flask is expected to produce the greatest volume of carbon dioxide during the first ten minutes?

**A** A

**B** B

**C** C

**D** D

**E** A and B must produce exactly the same amount

#### WORKED SOLUTION

Flask C contains boiled yeast, so its cells and enzymes will no longer function normally. Flask D lacks glucose.

Both A and B contain living yeast and glucose, but B is at the stated suitable higher temperature of 35°C, so respiration should proceed faster. Therefore **B** should produce the most carbon dioxide in the first ten minutes.

**B-16**

Thermoregulation

Answer **B**

#### QUESTION 16

A healthy person's core body temperature rises above its normal value.

Which response is expected?

**A** sweat production decreases; skin arterioles constrict

**B** sweat production increases; skin arterioles dilate

**C** sweat production increases; skin arterioles constrict

**D** sweat production decreases; skin arterioles dilate

**E** sweat production unchanged; skin arterioles constrict

#### WORKED SOLUTION

When body temperature is too high, sweat production increases and skin arterioles dilate. Both changes increase heat loss from the body.

**B-17**

Synapses

Answer **E**

**QUESTION 17**

Which statements about a synapse are correct?

1. Neurotransmitters are chemicals.
2. Neurotransmitter molecules can diffuse across the synaptic gap.
3. The electrical impulse itself passes directly across the gap as an electrical current.

**A** none

**B** 1 only

**C** 2 only

**D** 3 only

**E** 1 and 2 only

**F** 1 and 3 only

**G** 2 and 3 only

**H** 1, 2 and 3

**WORKED SOLUTION**

Neurotransmitters are chemical messengers. They are released from one neurone and diffuse across the synaptic gap.

The electrical impulse does not simply travel directly across the gap as an electrical current. Therefore **1 and 2 only**.

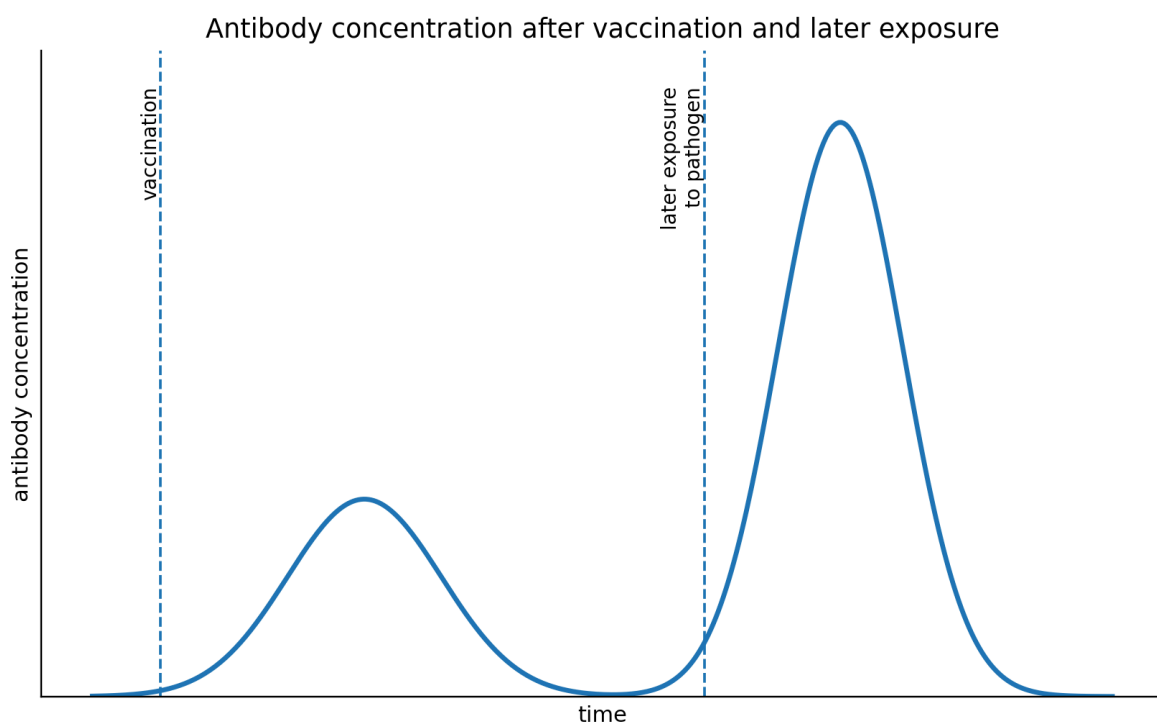
**B-18**

Vaccination

Answer E

**QUESTION 18**

The graph shows antibody concentration following vaccination and a later exposure to the same pathogen.



Which statements could explain the graph?

1. Vaccination can lead to the formation of memory cells.
2. Memory cells can allow a faster and larger antibody response during later exposure.
3. The vaccine must contain fully active pathogens capable of producing severe disease.

**A** none

**B** 1 only

**C** 2 only

**D** 3 only

**E** 1 and 2 only

**F** 1 and 3 only

**G** 2 and 3 only

**H** 1, 2 and 3

#### WORKED SOLUTION

Vaccination can generate memory cells. On later exposure to the same antigen, these cells allow a faster and larger antibody response.

A vaccine does not have to contain fully active disease-causing pathogens. Therefore **1 and 2 only**.

**B-19**

Antibiotic resistance

Answer **E**

#### QUESTION 19

A bacterial population contains a small number of bacteria that are resistant to antibiotic X. Antibiotic X is then used.

Which statements correctly describe natural selection in this population?

1. Resistant bacteria may have been present before the antibiotic was applied.
2. Susceptible bacteria are more likely to die, increasing the relative reproductive success of resistant bacteria.
3. The antibiotic deliberately causes every surviving bacterium to mutate to become resistant.

**A** none

**B** 1 only

**C** 2 only

**D** 3 only

**E** 1 and 2 only

**F** 1 and 3 only

**G** 2 and 3 only

**H** 1, 2 and 3

#### WORKED SOLUTION

Resistance-conferring variation can exist before treatment. The antibiotic then acts as a selective pressure: susceptible bacteria die more readily while resistant bacteria survive and reproduce.

The antibiotic does not deliberately cause every survivor to mutate. Therefore **1 and 2 only**.

**B-20**

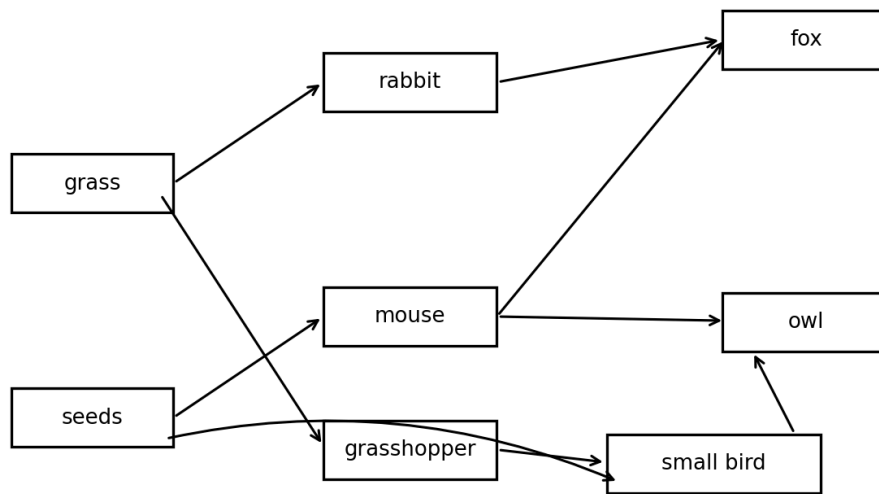
Food webs

Answer **E**

#### QUESTION 20

Arrows in the food web show transfer of biomass from food to consumer.

Arrows show transfer of biomass from food to consumer



Which organism can act as a **primary consumer** and a **secondary consumer**, and is also eaten by an owl?

- A** rabbit
- B** mouse
- C** fox
- D** grasshopper
- E** small bird

#### WORKED SOLUTION

The small bird eats seeds, so when doing so it is a primary consumer. It also eats grasshoppers, so it can act as a secondary consumer.

The food web also shows the small bird being eaten by the owl. Therefore the answer is **small bird**.

**QUESTION 21**

Five  $1.0 \text{ m}^2$  quadrats are placed randomly in a  $200 \text{ m}^2$  field.

The number of individuals of plant species X recorded is:

| quadrat | number of plants |
|---------|------------------|
| 1       | 4                |
| 2       | 6                |
| 3       | 5                |
| 4       | 3                |
| 5       | 7                |

Using these samples, what is the best estimate of the number of plants of species X in the whole field?

**A** 200

**B** 400

**C** 500

**D** 1000

**E** 5000

**WORKED SOLUTION**

The mean number per  $1.0 \text{ m}^2$  quadrat is:

$$(4 + 6 + 5 + 3 + 7) / 5 = 25 / 5 = 5$$

Estimated density =  $5 \text{ plants m}^{-2}$ .

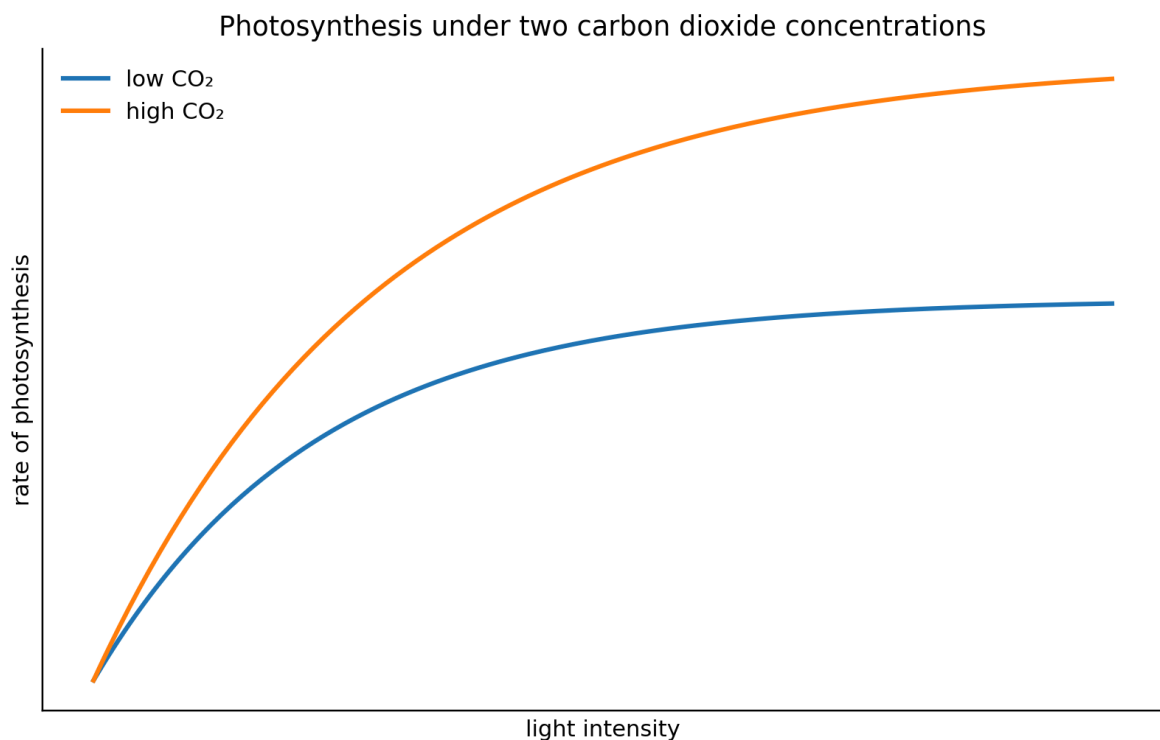
$$5 \times 200 = 1000 \text{ plants}$$

**B-22**

Limiting factors in photosynthesis

**Answer A****QUESTION 22**

The graph shows the effect of light intensity under two carbon dioxide concentrations.



Which conclusion is best supported?

- A** At high light intensity, carbon dioxide concentration can limit the rate of photosynthesis.
- B** At low carbon dioxide concentration, light intensity can never be limiting.
- C** Increasing light intensity indefinitely must always increase photosynthesis.
- D** Carbon dioxide concentration has no effect on photosynthesis.

- E** Photosynthesis is independent of light intensity.

#### WORKED SOLUTION

At high light intensity, the low-CO<sub>2</sub> curve levels off below the high-CO<sub>2</sub> curve. Increasing carbon dioxide therefore increases photosynthesis under those conditions.

This supports the conclusion that carbon dioxide concentration can be a limiting factor at high light intensity.

**B-23**

Plant transport

Answer **B**

#### QUESTION 23

Which row correctly describes xylem and phloem?

- A** xylem transports dissolved sugars from leaves; phloem transports water to leaves
- B** xylem transports water and mineral ions mainly upwards; phloem transports dissolved sugars from leaves to other parts of the plant
- C** xylem transports sugars only downwards; phloem transports water only upwards
- D** xylem consists entirely of living cells; phloem consists entirely of dead cells
- E** xylem transports water by active transport along the vessel; phloem transports only mineral ions

#### WORKED SOLUTION

Xylem transports water and mineral ions mainly from the roots upwards through the plant.

Phloem transports dissolved sugars produced in leaves to other parts of the plant where they are used or stored.

**B-24**

Transpiration

Answer C

**QUESTION 24**

Identical leafy shoots are placed under different environmental conditions.

| experiment | temperature | humidity | air movement |
|------------|-------------|----------|--------------|
| A          | low         | high     | still        |
| B          | high        | low      | still        |
| C          | high        | low      | windy        |
| D          | low         | high     | windy        |

Which experiment would normally be expected to have the **highest rate of transpiration**?

**A** A

**B** B

**C** C

**D** D

**E** All four should have equal rates

**WORKED SOLUTION**

Transpiration generally increases with higher temperature, lower humidity and greater air movement.

Experiment C combines all three: **high temperature, low humidity and wind**. Therefore C should have the highest transpiration rate.

**B-25**

Human reproduction

Answer **B**

**QUESTION 25**

Which row correctly identifies the usual site of fertilisation and the usual site at which an embryo implants?

**A** fertilisation: ovary; implantation: oviduct

**B** fertilisation: oviduct; implantation: uterus

**C** fertilisation: uterus; implantation: ovary

**D** fertilisation: vagina; implantation: uterus

**E** fertilisation: uterus; implantation: oviduct

**WORKED SOLUTION**

In humans, fertilisation normally occurs in an **oviduct**. The developing embryo then reaches the uterus and implants in the **uterine lining**.

**B-26**

Variation

Answer **H**

**QUESTION 26**

Which statements are correct?

1. Human height is an example of continuous variation.
2. Human ABO blood group is an example of discontinuous variation.

3. Environmental factors such as nutrition can contribute to variation in human height.

**A** none

**B** 1 only

**C** 2 only

**D** 3 only

**E** 1 and 2 only

**F** 1 and 3 only

**G** 2 and 3 only

**H** 1, 2 and 3

#### WORKED SOLUTION

Human height can take many values across a range, so it is continuous variation. ABO blood groups form distinct categories and are therefore discontinuous.

Height is influenced by both genetic and environmental factors, including nutrition. Therefore all three statements are correct.

**B-27**

Competition

Answer **B**

#### QUESTION 27

Two plant species are grown either separately or together for the same length of time. Assume equal starting biomasses of each species are used in the relevant treatments, and all other environmental conditions are kept the same.

| treatment                 | mean final biomass of species<br>A / g | mean final biomass of species<br>B / g |
|---------------------------|----------------------------------------|----------------------------------------|
| A grown alone             | 24                                     | —                                      |
| B grown alone             | —                                      | 20                                     |
| A and B grown<br>together | 14                                     | 12                                     |

Which conclusion is best supported by these data?

- A** Species A and B do not compete.
- B** Growing together reduces the growth of both species, which is consistent with competition between them.
- C** Species A competes only for light.
- D** Species B will always become extinct when grown with A.
- E** The experiment proves that water is the resource for which the species compete.

#### WORKED SOLUTION

When grown alone, species A reaches 24 g and species B reaches 20 g. When grown together, both have lower final biomass: 14 g and 12 g respectively.

This is consistent with interspecific competition. However, the data do not identify the particular resource being competed for and do not show that either species must become extinct.

**Compilation record:** Level 0 uses Engineering Mock Test 1 for Mathematics/Physics and the corresponding Easy Biology/Chemistry source sets.

This solution book is ready to support the later construction of the matching full test.